VERA

Vulnerability Evidence and Reasoning Agent

Evidence-based security analysis with a local model and an inspectable workflow.

Workflow
One stateful LangGraph

Routes
Five acquisition paths

Model
Local and bounded

State
SQLite + Kuzu

The model, API and explorers run on the local machine.

Alerts and unsupported claims

01 · signal

A port scan is reconnaissance

It shows scanning activity, but not whether a host was compromised.

02 · fluency

Fluent output without support

A model answer may be fluent while its citations are missing or irrelevant.

03 · proof

Claims are checked against evidence

The validator checks identifiers, numbers and quotations against fetched evidence.

Assessments report evidence, confidence and completeness.

Detect	Retrieve	Validate	Review	Report
signal from events	local + graph evidence	citations checked	human review for high risk	assessment, limits, sources

What VERA is

VERA is a teaching prototype for evidence-based security analysis: one LangGraph workflow over curated knowledge, local retrieval, a knowledge graph and event rules.

01

Quoted evidence

Supported facts quote non-memory evidence verbatim.

02

Bounded model use

Deterministic checks verify which statements are supported.

03

Local execution

The model, API and explorers run locally.

04

Visible workflow

The UI exposes routes, evidence, warnings and reviews.

In scope: one workflow, five routes, a local model, a trace.

Out of scope: agent swarms, cloud dependency, asset inventory.

Architecture and repository layout

Modules are separated by typed contracts.

domain
Request, evidence, claim and result contracts.

routing
Intent rules; optional LLM disambiguation.

retrieval
Ranking, embeddings, corpus sync.

sources
Read-only CVE.org, NVD, KEV, EPSS, OSV.

stream
Events, PCAP, detectors, statistics.

providers
Deterministic, OpenAI-compatible, Ollama, Gemini.

reasoning
Citation validation, repair, reflection.

memory
SQLite history, memories, review ledger.

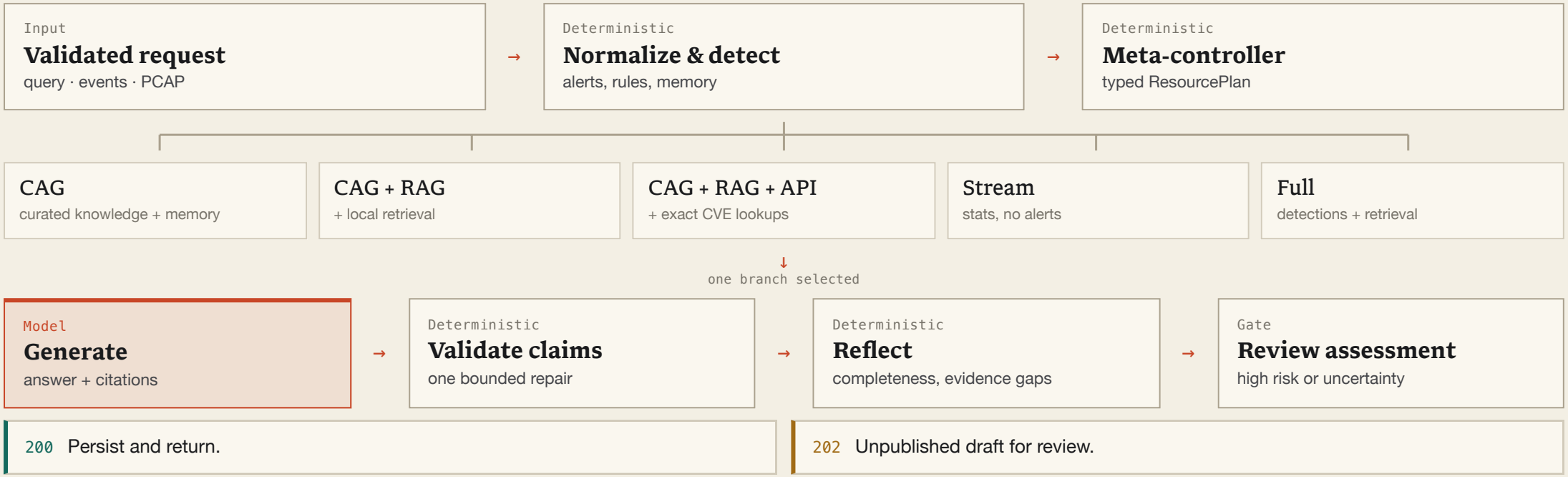
workflow
LangGraph topology and wiring.

interfaces
FastAPI API and Typer CLI.

```
src/vera/ → domain · workflow · retrieval · sources · stream · providers · reasoning · memory · interfaces · data  |  finetuning/ · presentation/ · scripts/
```

Workflow and acquisition paths

deterministic model involvement One resource plan selects one branch.



Context acquisition always runs between routing and generation.

Routing: four independent flags

Four independent flags. The `Route` label is derived; mixed requests combine resources.

Independent flags

CAG	Curated knowledge + memory.
RAG	Local corpus; optional Kuzu.
TOOL	Read-only CVE lookups.
STREAM	Events and detectors.

CAG-only applies when curated knowledge covers the topic. Live lookups need explicitly enabled sources.

Route	CAG	RAG	Tool	Stream	When
CAG	●	○	○	○	Conceptual question with curated coverage.
CAG + RAG	●	●	○	○	Needs documentary sources.
CAG + RAG + API	●	●	●	○	CVE IDs present; live sources enabled.
Stream	○	○	○	●	Events without alerts.
Full	●	●	●	●	Active alerts, e.g. `NET-PORT-SCAN`.

> One request can combine stream detection, retrieval and exact API lookups.

Model role and deterministic checks

Deterministic layer

- ✓ Routing and event detection
- ✓ BM25 + embedding ranking, provenance rerank
- ✓ Citation validation and lexical grounding
- ✓ Reflection on completeness and gaps
- ✓ Human-review triggers
- ✓ Graph projection and persistence

Without a model, the **deterministic** provider still runs the workflow.

Model tasks

- Resolve routing ambiguity (optional)
- Generate the answer and its citations
- Make one correction after an invalid claim
- Choose a ReAct action (optional, bounded)

No graph writes, arbitrary queries, URLs or commands. Memory cannot establish a fact.

Call budget

2 calls by default; `VERA_MAX_REACT_STEPS + 3` with ReAct.

Providers

Deterministic, OpenAI-compatible, Ollama, Unsloth, Gemini.

Explicit degradation

`provider_fallback`, `warnings`, model metadata.

How claims are validated

Claim types

supported_fact
A verbatim quotation from non-memory evidence.

system_inference
A structured mapping; still validated.

unavailable
Missing evidence, stated explicitly.

simulated
Synthetic data, labelled end to end.

What the validator checks

01 **Registry integrity**
Unknown IDs cannot support a fact.

02 **Quotation presence**
The quote must appear in the evidence.

03 **Coverage**
Terms, IDs, numbers and negation.

04 **Contradictions**
Irrelevant citations and simple contradictions are rejected.

05 **Bounded repair**
One retry, then an explicit inference.

> The validator performs **lexical checks** against grounded evidence. Coverage is computed from grounded results.

Hybrid retrieval and ranking

Hybrid score



Normalized scores. Exact IDs stay single tokens.

Above the relevance threshold

Candidates	Rerank	Dedupe	Top-k
bounded set	provenance · date · novelty	near-identical documents	selected evidence

- **Hash embeddings** by default; Sentence Transformers optional.
- **Ranking order:** live records, then synchronized feeds, then snapshots.
- Undated guidance gets a **neutral freshness** score.

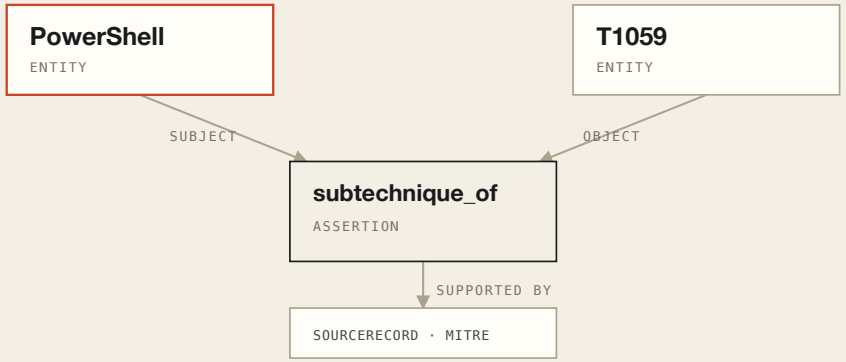
Quality signals exposed per result

Field	Meaning
`retrieval_score`	Original hybrid score
`quality_score`	Source quality
`freshness_score`	Freshness; neutral when undated
`novelty_score`	Novelty against the rest
`selection_score`	Final selection score

> **Explicit sync.** Allowlisted ATT&CK STIX and CISA KEV only; validated and staged before publishing.

Scores rank evidence; they do not verify source accuracy.

Knowledge graph: Kuzu



120
entities

112
supported assertions

ATT&CK 17.1 | CWE 4.20 | CAPEC 3.9 | KEV 2026.10.08 | OWASP 2021 | NIST

Node	Meaning
`Entity`	CVE, CWE, CAPEC, technique, tactic, mitigation, group, software, concept
`Assertion`	A sourced statement: subject + predicate + object
`SourceRecord`	Immutable evidence version with title, URL, status and metadata
`Run · Alert · Event`	Technical projection of the analysis history

`subtechnique_of` | `has_weakness` | `part_of` | `mitigates` | `uses` | `child_of` | `exploits_weakness` | `covers`

Traversal bounds

Hops
one or two logical hops; default one

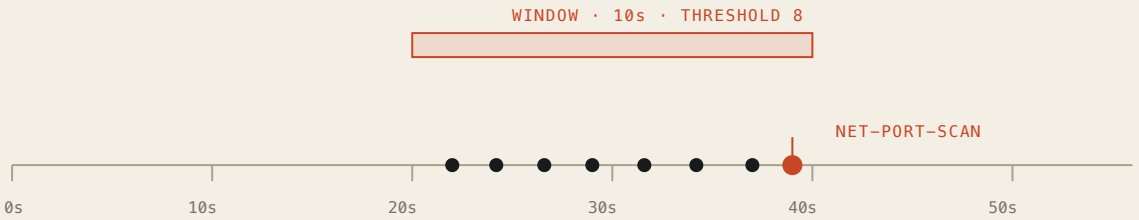
Results
12 per frontier, 8 identifiers max

Timeout
3,000 ms per query

Queries
fixed Cypher templates, bound inputs

> No arbitrary queries or graph writes. If Kuzu is missing, retrieval continues with **warnings**.

Stream detection and PCAP input



Alerts expose count, threshold, timestamps, duration and confidence.

Bundled synthetic sample

```
vera analyze "Analyze this traffic" \
  --events src/vera/data/samples/port-scan.jsonl \
  --simulated --offline --json
```

`benign-web` → `STREAM`, no alerts. `port-scan` → `FULL`, `NET-PORT-SCAN`.

Sliding windows

Port scans, auth failures and SYN bursts use configurable windows.

Event-local signatures

HTTP payload signatures stay event-local.

Restricted PCAP

PCAP is parsed from an argument list without a shell. Packet bytes stay outside model context.

Input rules

- ✓ Events via API or `--events`; raw logs in `query` are ignored.
- ✓ PCAP needs the CLI and `tshark`.
- ✓ `--simulated` labels synthetic data; `--offline` disables live lookups.

When human review is required

Deterministic review triggers

- High / critical alert
- Unsupported claim
- Missing usable evidence
- Out-of-scope conflict

Deterministic

Reflect

completeness and evidence gaps

→

Gate

Review assessment

ordinary path returns 200

200 · approve

Saved report

Persisted as a completed run.

200 · reject

Closed review

Nothing published; feedback required.

202 · revise

Pauses again

One model run, then pauses. Max two cycles.

Unpublished: drafts never appear in `/v1/runs`, history, memory or graph projection.

Local decisions: loopback and Origin checks; no identities. The CLI uses the same review gate.

Optional ReAct mode

Optional ReAct loop · `VERA_REACT_ENABLED=true`

Model decision
react_decide
typed action + short operational reason

↓

Deterministic tool
react_execute
local search · exact CVE lookup

↓

Observation
New evidence in the registry
back to decide

↺
until finish or budget

Exit
finish → Generate
generation resumes

Budget

2 calls default. ReAct: `steps + 3`. 3 steps → 6 calls; 10 → 13.

- Local context is tried first; tools and full mode defer live lookups.
- Typed decisions: corpus search and exact CVE lookups.
- Every decision needs a **short reason**.
- Reasons and outcomes land in the persisted trace.
- The loop cannot spend the two generation calls.

The trace records decisions and outcomes, not hidden reasoning.

No web search, arbitrary URLs or commands.

Safety boundaries

Shell-free PCAP
Argument lists for PCAP; fixed read-only HTTP for tools.

Packet data stays local
Packet bytes are parsed locally; evidence is treated as untrusted data.

Bounded work
Input, PCAP, model calls, tool calls and timeouts have limits.

Verbatim evidence
Unknown IDs cannot support a fact; quotes are required.

Explicit missing data
Missing data is recorded as **unavailable** evidence.

Loopback-only server
Loopback only; foreign clients and Origins are rejected.

Missing evidence is recorded as unavailable and remains citable.

Design rule

Persistence and recovery

Written by a completed run

01 **Run history**
Requests, evidence, reflection.

02 **Semantic memory**
Supported findings used only as context.

03 **Episodic memory**
Request, route, outcome.

04 **Checkpoints + ledger**
Drafts and versioned review decisions.

05 **Graph projection**
Committed with history; idempotent retries.

Recovery, in order

- `interrupt()` after reflection.
- `Command(resume=...)` continues the graph.
- Compare-and-set decisions with idempotency keys.
- Startup reconciles; missing state fails closed.

`review`
Max two revision cycles.

`conflict`
Stale version → `409`.

`pruning`
Terminal threads pruned.

`server`
One process per database.

CLI: `vera reviews` · `vera review <run_id> --approve`

> Drafts **never** appear in `/v1/runs`, history, memory or graph projection.

Interfaces

HTTP API `/docs`	`POST /v1/analyze` → report (`200`) or draft (`202`). Read runs, topology, reviews and health.	`/v1/analyze · /v1/runs · /v1/graph · /v1/reviews`
Typer CLI `vera`	The CLI exposes the same review gate without a server: analyze, serve, history, evaluate, import, review.	`--events · --pcap · --offline · --simulated · --json`
Workflow Explorer `/graph`	Compiled topology, previews, recorded runs, contracts and Mermaid. Previews never call a model.	`topology · preview · execution`
Knowledge Graph Explorer `/knowledge-graph`	Search by name, alias or ID; inspect neighbours, predicates, sources and provenance.	`entities · predicates · sources`

All components are served locally; there is no external renderer.

Local model setup

NVIDIA · 6 GB VRAM

Qwen3.5-4B

GGUF Q4_K_M via LM Studio CUDA, served as `vera-qwen3.5-4b`. 8,192 context, thinking off.

Apple silicon

Qwen3.5-4B MTP

GGUF Q6_K via `llama-server` with MTP. About **70% faster** decoding, same local quality.

Model files are **not** in the repository.

Verify the connection

- ✓ Model ID matches.
- ✓ `provider_fallback=false`: model answered.
- ✓ Nonempty `cited_evidence_ids`: graph evidence cited.
- ✓ `/health` is configuration, not inference.

Configuration

```
VERA_LLM_PROVIDER=openai
VERA_LLM_MODEL=Qwen3.5-4B
VERA_LLM_BASE_URL=http://127.0.0.1:1234/v1
VERA_KG_ENABLED=true
VERA_RETRIEVAL_K=3
VERA_MEMORY_K=1
VERA_REACT_ENABLED=false
VERA_LIVE_SOURCES=false
```



No LLM? `VERA_LLM_PROVIDER=deterministic` still runs.

Separate processes: `vera serve` does not start Qwen.

Fine-tuning a local adapter

3,939
training examples · 11 tasks

0.81
epoch · 160 → 400 updates

8
LoRA rank · alpha 16

0.00044
final validation loss

Training run

- ✓ **Base:** Qwen3.5-4B, revision `851bf6e8`, FP32 on two T4.
- ✓ **Adapter:** rank 8, alpha 16, seven projection modules.
- ✓ **Data:** 3,939 train / 528 validation / 522 test; 2,560-token budget.
- ✓ **Cost:** 8 h 09 min for 240 updates, plus ~35 min validation.
- ✓ **Artifact:** 42 MB adapter; 21 MB F16 LoRA GGUF for local serving.

What it fixes

Valid JSON, scope, missing-evidence and injection behavior on the prepared probes.

What it does not fix

A 0.81-epoch run over templated targets cannot add capabilities the dataset never shows.

Deployment status

Experimental. Targets are synthetic and unreviewed; the contract forbids training and deployment.

Validation loss is already saturated: another epoch would add memorization, not capability.

v4 dataset: closing the measured gaps

6,140
examples · 4,863 / 622 / 655

0–3
alert counts in every split

2,462
longest input token count

2560
max_length · no truncation

Gap → v4 response

Prompt drift	current system and generation prompts
One-sided scope	in-domain, unavailable, social, out-of-scope
Alerts only 0/1	counts 2 and 3 plus correlated replays
No multi-turn	follow-ups with resolved subject
No language variants	IT·ES·DE·FR → configured English
Entity drift	verbatim quotes + entity bindings

Why 2,560

2,048 drops 80 multi-alert rows and 2,304 still truncates 48; 2,560 keeps alert counts 2–3.

Still unauthorized

Synthetic, unreviewed targets. `training_authorized` and `human_reviewed` stay false.

The audit checks split, source, entity and verbatim-fact leakage, replay targets against the real detectors and the conversational contract. It does not prove the model improves.

English-only evaluation

109/110
adapter strict passes · base 43/110

33/40
adapter perturbed · base 23/40

89/90
adapter on unseen targets · base 39/90

110-case Kaggle pack

Metric	Base	Adapter
Valid schema	88.2%	**100%**
Correct scope	82.7%	**100%**
Status recall	0.670	**1.000**
Citation failures	28.9%	**0.9%**
Memory violations	10	**0**
Mean output tokens	482	**218**

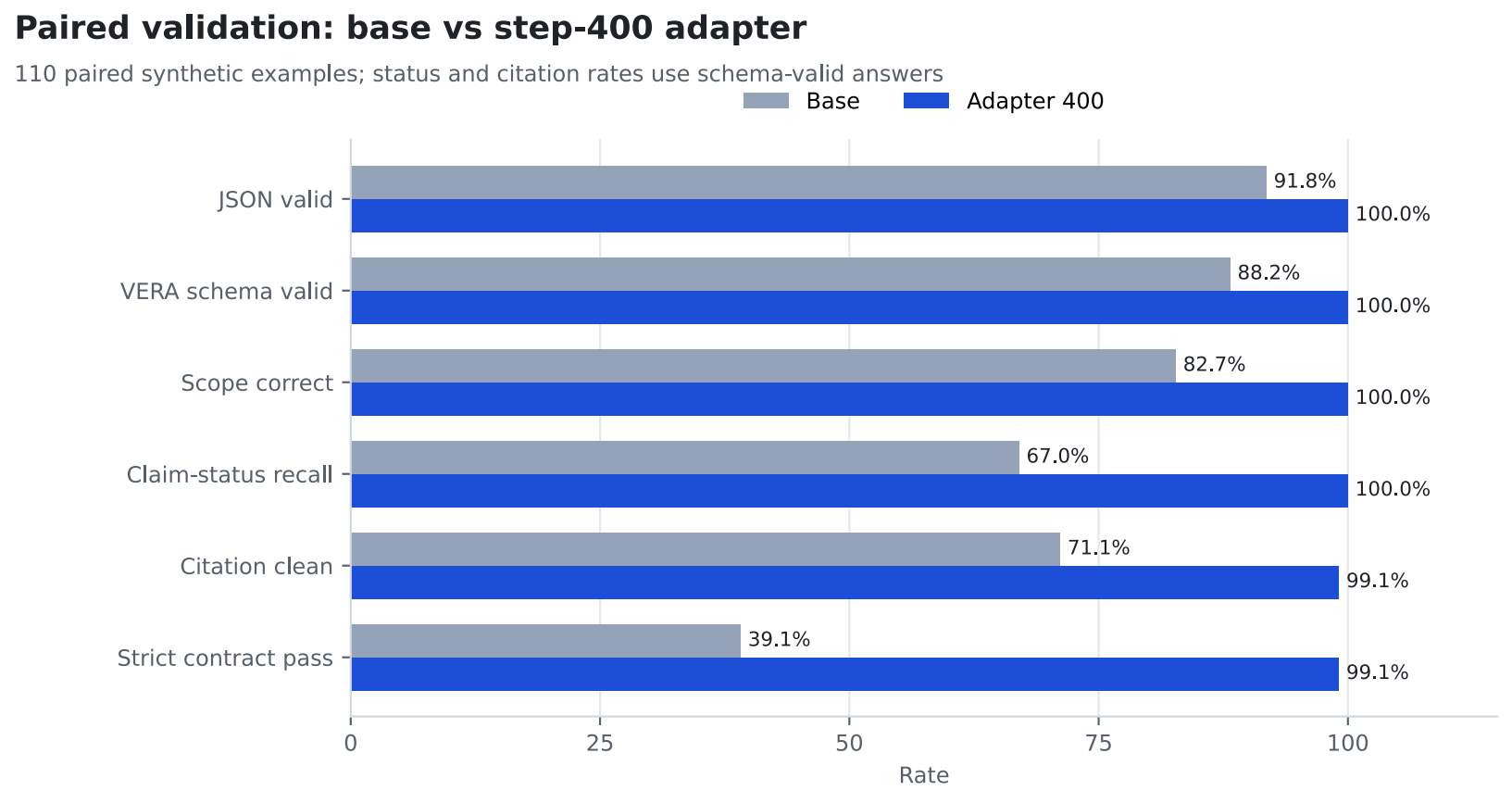
40-case perturbed set

Family	Base	Adapter
scope · unavailable	5/12	**12/12**
injection · multiturn	9/10	**10/10**
language	3/4	**4/4**
entity	**6/6**	5/6
alerts	0/8	2/8

Paired tests: **p = 0.0063** for the rule check, **p = 0.000061** with grounding.

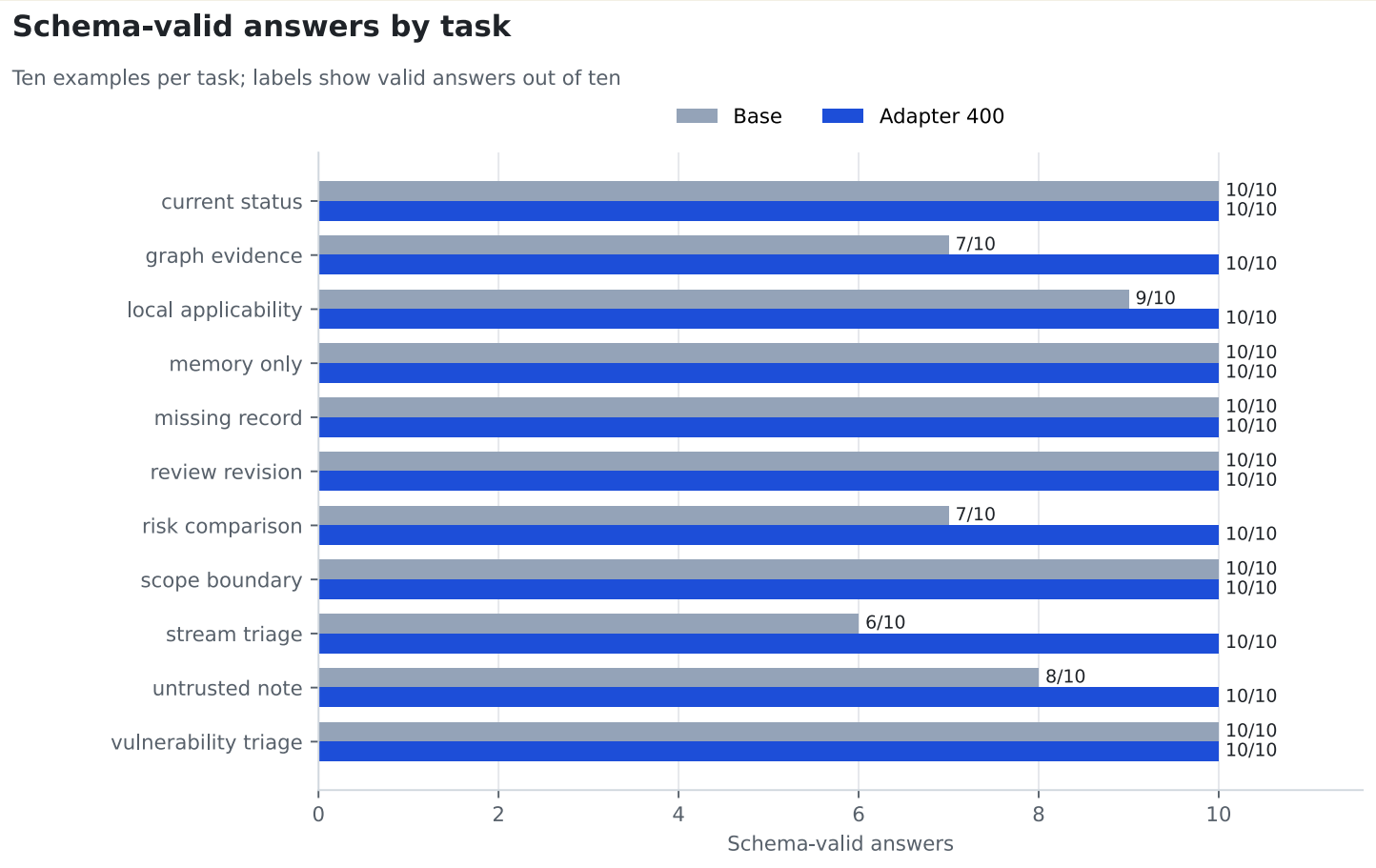
Both packs overlap training. They show retention and format robustness, not generalization; alert counts above one and long evidence quotes remain dataset defects.

Paired validation: base vs adapter



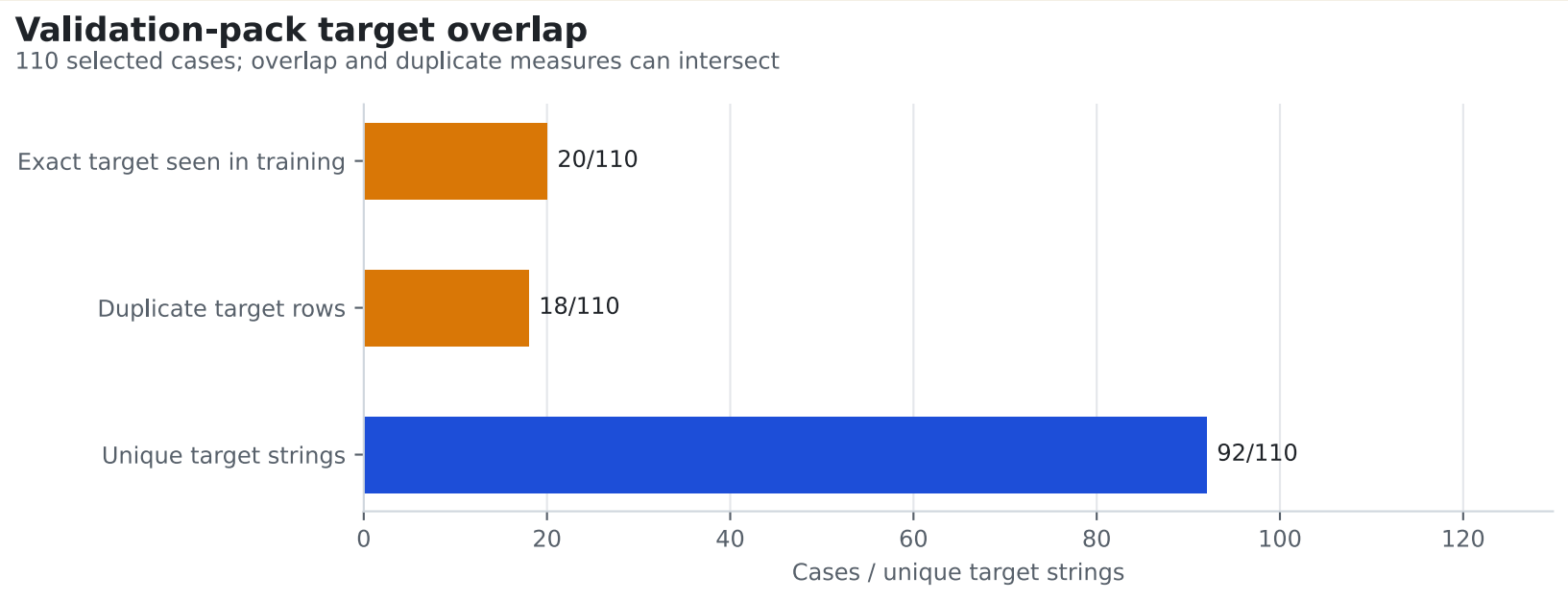
The adapter is at or near the ceiling on every contract metric; the base loses most on strict passes and claim-status recall.

Schema validity by task



All eleven tasks reach 10/10 with the adapter; the base drops on graph evidence, risk comparison, stream triage and untrusted notes.

Validation-pack composition



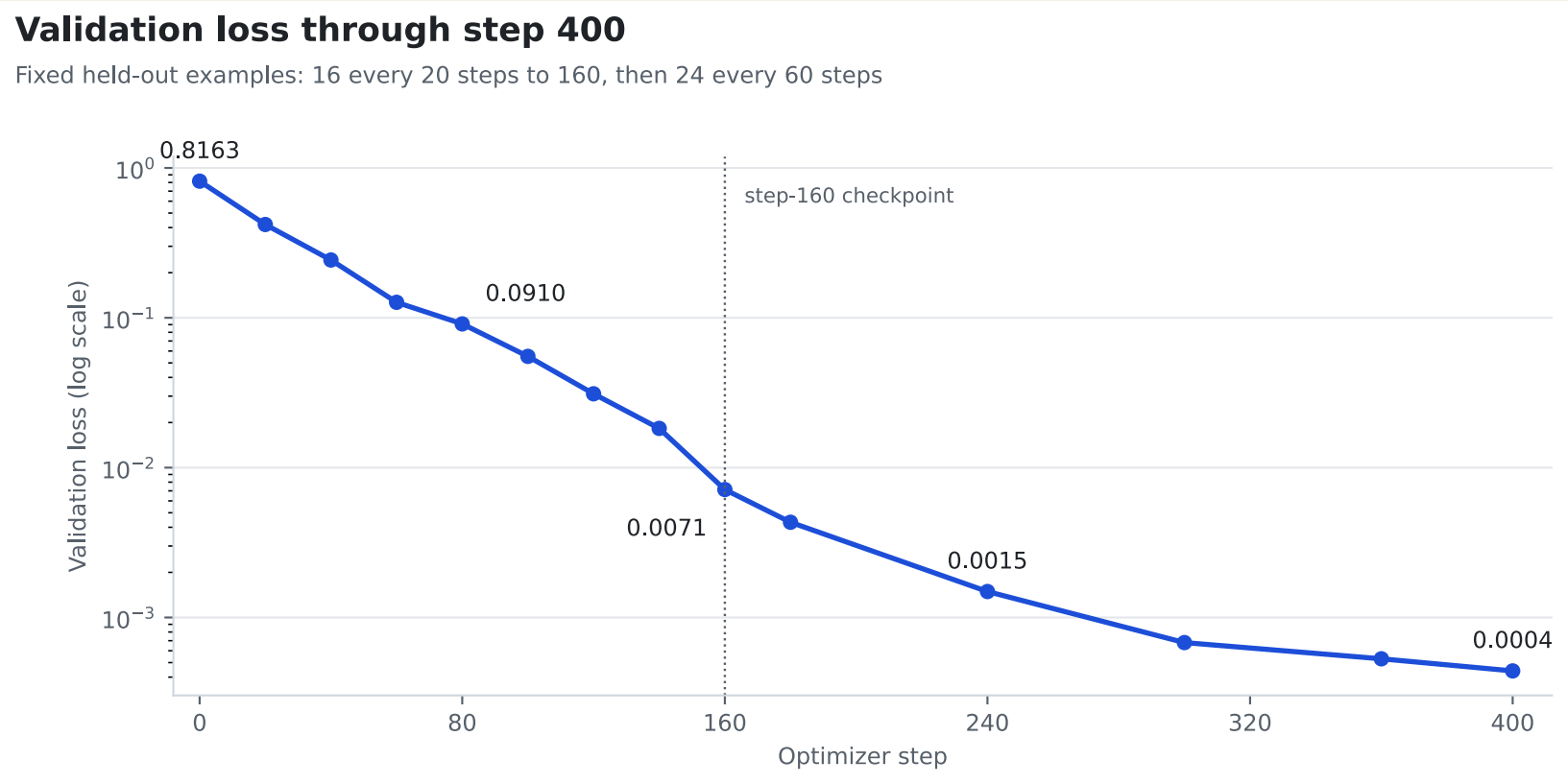
20
exact targets also occur in training

18
rows duplicate another target

92
unique target strings out of 110

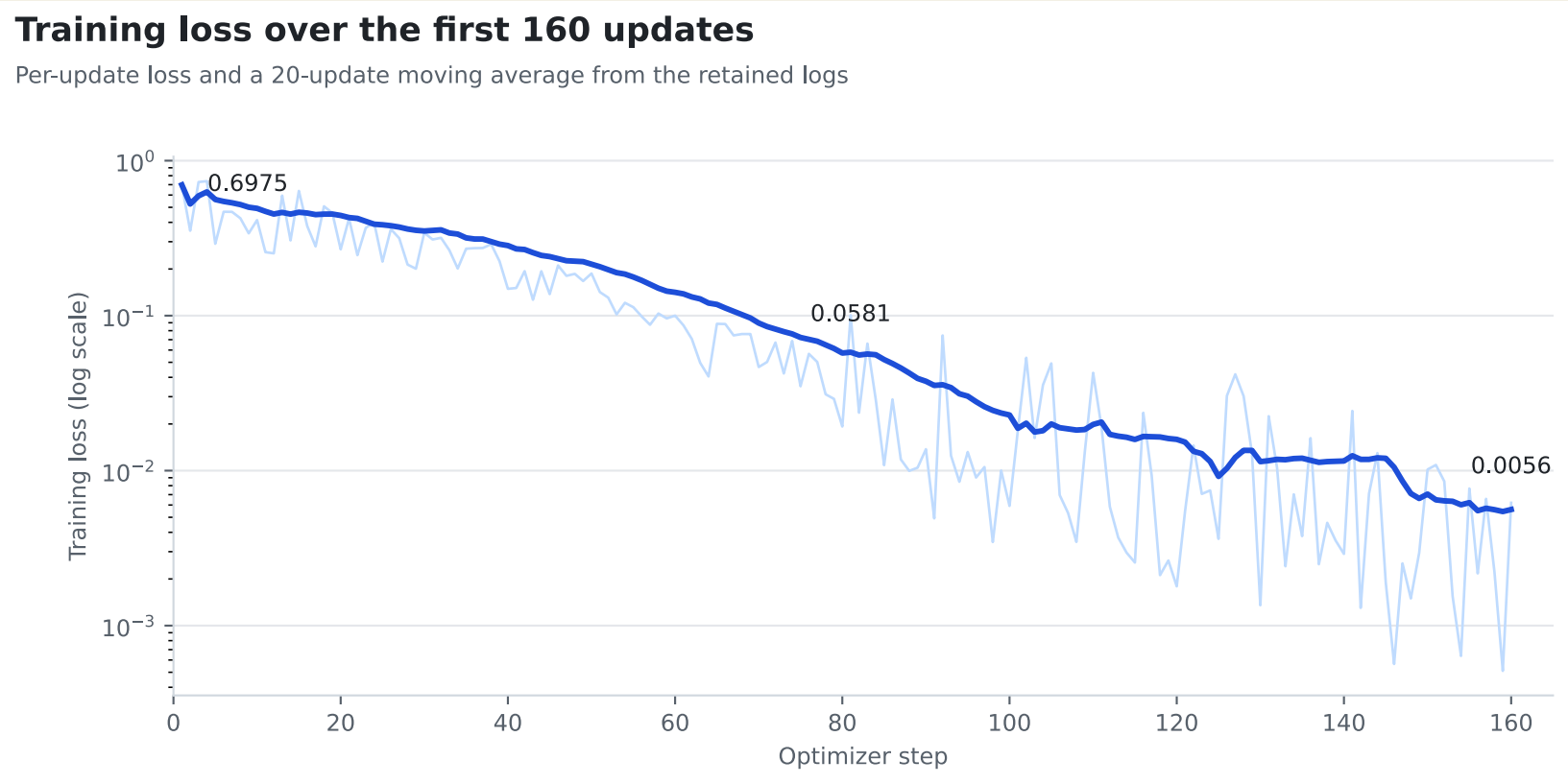
The pack measures retention and format robustness, not generalization.

Validation loss through step 400



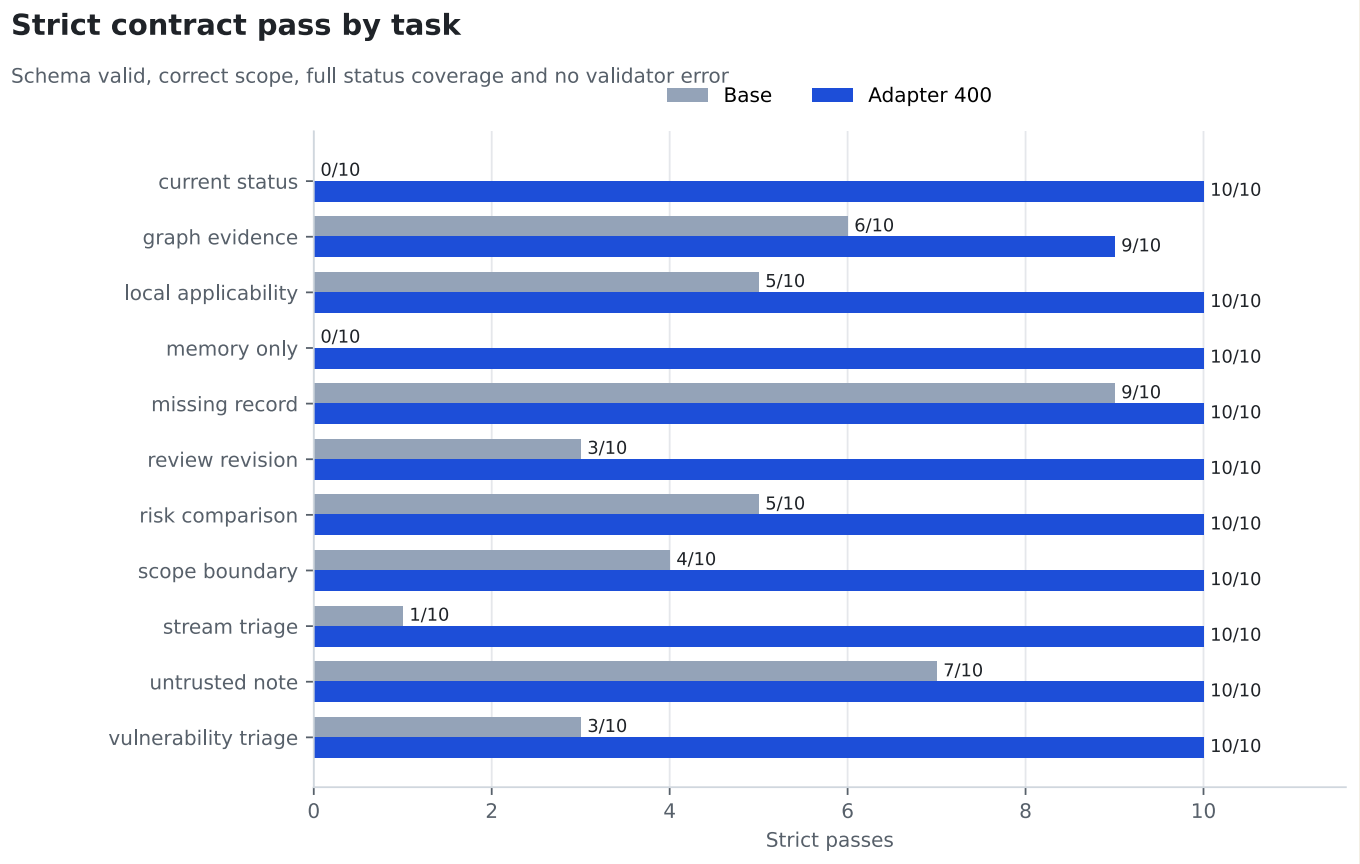
Loss falls from 0.816 to 0.00044. After step 240 each validation pass buys less than the previous one, so another epoch would add memorization, not skill.

Training loss, first 160 updates



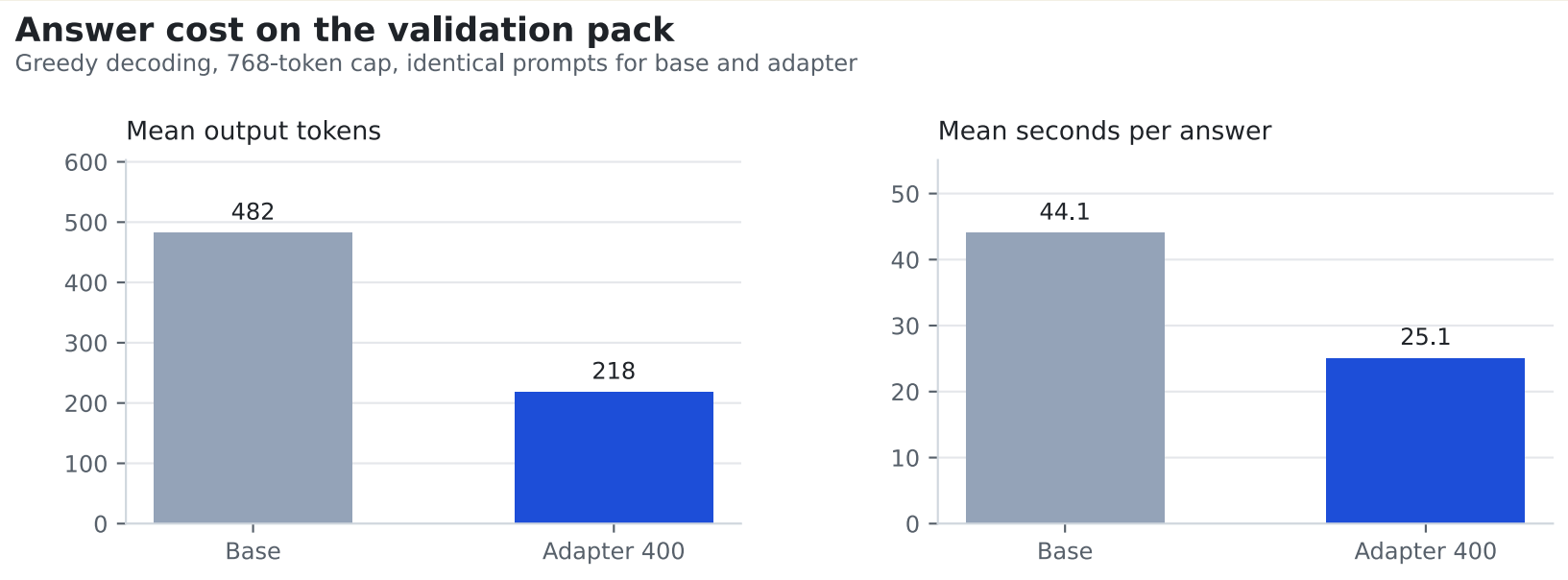
Per-update loss is noisy; the 20-update mean drops from 0.70 to 0.006, so the apparent smoothness of the validation curve is not an artefact of a single run.

Strict contract pass by task



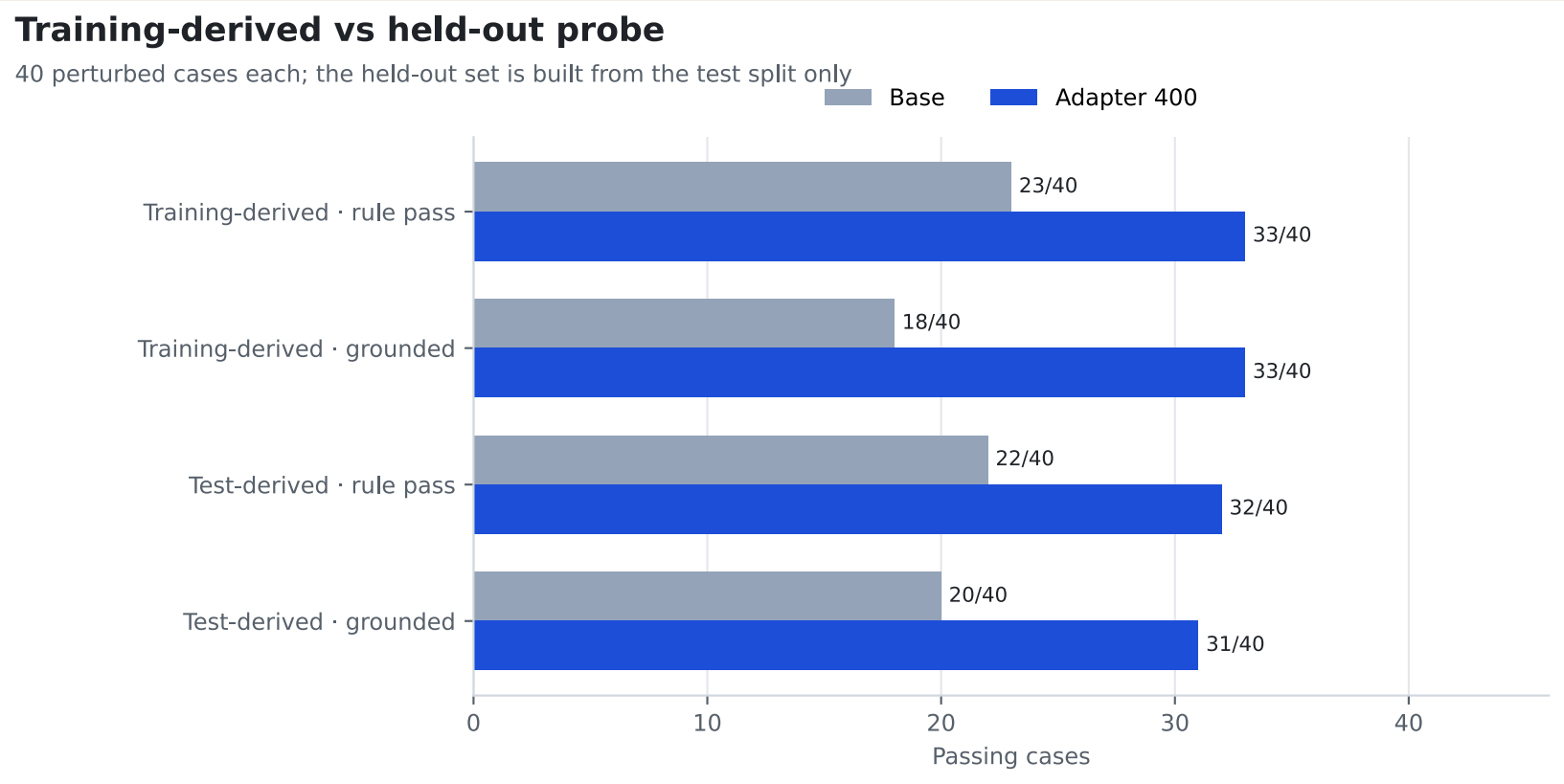
The adapter reaches 10/10 on ten tasks and 9/10 on graph evidence. The base collapses on current status, memory-only and stream triage, where scope and status coverage fail together.

Answer cost



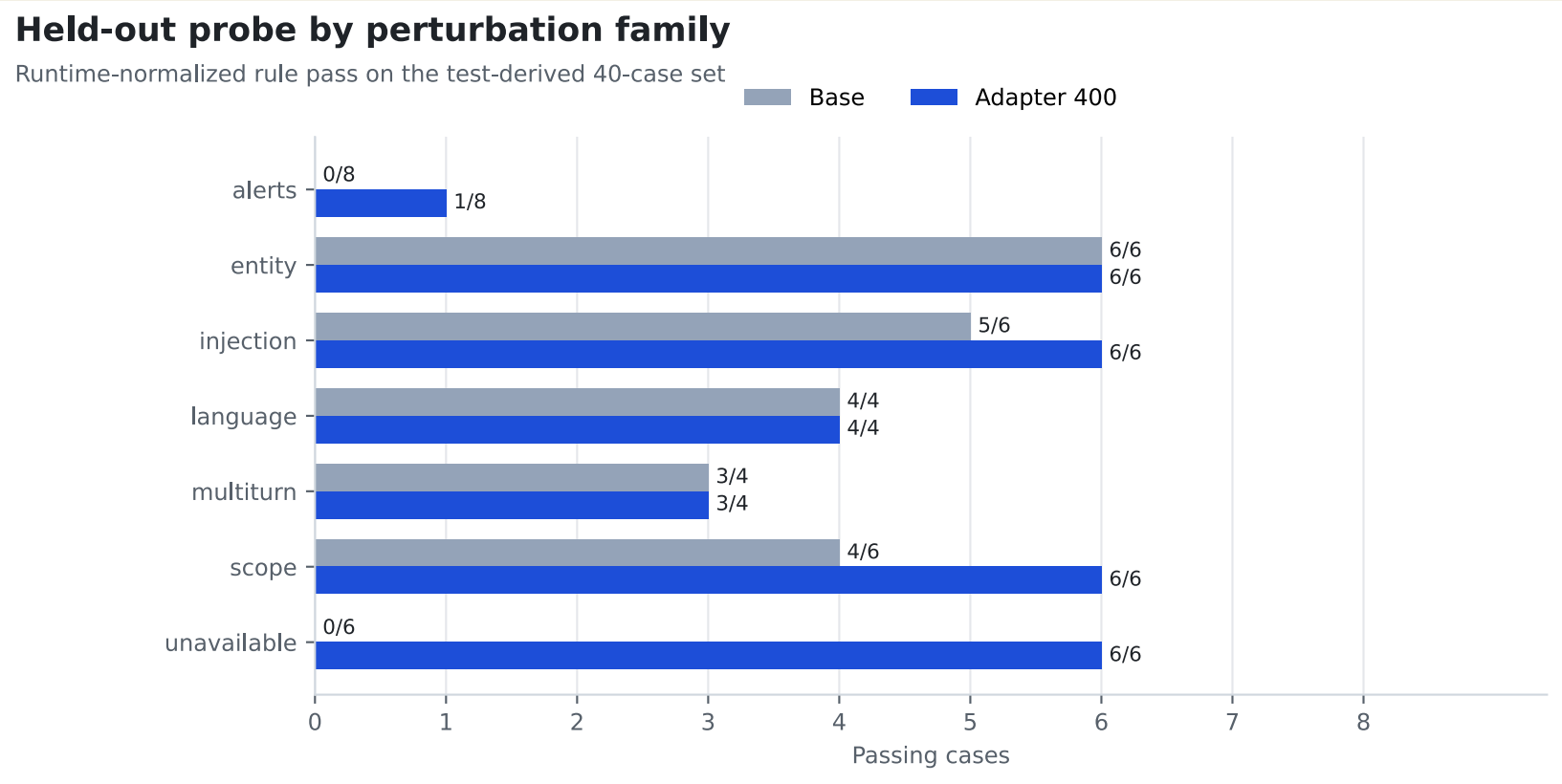
The adapter emits less than half the tokens and answers in about 57% of the base latency, on the same prompts and the same 768-token cap.

Held-out probe: training vs test split



On 40 cases built from the held-out test split the adapter still passes 32/40 versus 22/40 for the base, and 31/40 versus 20/40 with grounding. On the 220-case test pack it reaches 219/220 strict passes versus 86/220.

Held-out probe by family



Missing evidence transfers fully. Scope does not: both models refuse in-domain tool questions, and the adapter refuses more than the base. Alert counting stays 1/8 because training only contains zero or one configured signal.

Scope probe: refusal in both directions

13 cases · same pinned Q6_K model

Family	Base refused	Adapter refused
In-domain (30)	6	**19**
Unrelated (6)	6/6	6/6
Social (4)	4/4	4/4

All four social messages are wrong for both: the contract requires `out_of_scope=false` and no claims.

In-domain refusals

Both	PowerShell, SQL injection, XSS, phishing
Base only	MITRE ATT&CK, hardening
Adapter only	13 more, e.g. Nmap, Kerberoasting, Log4Shell, MFA
Both answer	CWE-89, CVE-2021-44228, firewall, T1059.001

16 of 19 adapter refusals have the expected record in the corpus; 3 are coverage gaps. Switching embeddings moved retrieval hits 22→24/30 and changed zero refusal decisions on the 10 key cases.

Directional result: 40 cases, one configuration. It qualifies the near-100% scope figure, which mostly measures rejection of unrelated requests.

Code size and validation

8,653
lines of Python source

119
source modules

11
fine-tuning tasks

23
presentation slides

Validation layers

01 **Lint**
`ruff check .` in CI on Python 3.11 and 3.13.

02 **Deterministic evaluation**
`vera evaluate --json` checks routing and grounding offline.

03 **Adapter evaluation**
40-case perturbed set plus 110/220-case validation packs.

No tracked test suite on `main`; the deterministic provider's English strings are hardcoded, so it does not test the language contract.

Development commands

```
E="uv run --extra dev --extra knowledge-graph"

$E ruff check .
$E vera evaluate --json
$E vera corpus-sync
$E vera kg-build
```

> Offline evaluation disables review to measure routing and grounding.

Dependencies

Runtime

- **Python 3.11+** · packages managed with `uv`
- **LangGraph** `≥0.2.60,<2` — workflow orchestration
- **langgraph-checkpoint-sqlite** `≥3.1,<4` — checkpoints
- **FastAPI** `≥0.115,<1` · **uvicorn** `≥0.34,<1`
- **Pydantic** `≥2.10,<3` · **pydantic-settings** `≥2.7,<3`
- **Typer** `≥0.15,<1` — CLI
- **httpx** `≥0.27,<1` — HTTP client

Versions are declared in `pyproject.toml` and locked by `uv.lock`.

Optional extras

- **kuzu** `==0.11.3` — `knowledge-graph` extra, pinned
- **sentence-transformers** `≥3.3,<6` — `embeddings` extra
- **ruff** `≥0.9,<1` — `dev` extra and CI lint gate
- **matplotlib** `≥3.9,<4` — `analysis` extra for charts

> Kuzu's upstream repository is **archived**: the dependency is pinned to 0.11.3 and the integration requires no extensions.

Not included: cloud SDK, external vector database, graph server, frontend build step.

Getting started

1 **Install**

```
uv sync --extra dev --extra knowledge-graph
cp -n .env.example .env
```

2 **Start a local model on port 1234**

```
# Bionic / LM Studio: load Qwen3.5-4B, serve :1234
curl -fsS http://127.0.0.1:1234/v1/models
```

Or, on Linux/NVIDIA: `bash scripts/start-local-model.sh`.

3 **Configure**

Set provider, exact model ID, base URL and `VERA_KG_ENABLED=true`. Restart after `.env` changes.

4 **Build the graph and serve**

```
uv run --extra knowledge-graph vera kg-build
uv run --extra knowledge-graph vera kg-status
uv run --extra knowledge-graph vera serve --open-browser
```

5 **Ask the first question**

```
curl -fsS http://127.0.0.1:8000/v1/analyze \
  -H 'Content-Type: application/json' \
  -d '{"query":"Explain PowerShell. Give a brief sourced claim."}'
```

Model server on `:1234`, VERA on `:8000`. `vera serve` does not start Qwen.

Where to look

`/docs · Swagger` `/graph · workflow`

`/knowledge-graph · domain` `/health · configuration`

No model? `VERA_LLM_PROVIDER=deterministic` still runs the workflow.

Check `metadata.model` and `provider_fallback=false`.

Limitations

No asset inventory
No version matching; no automatic CVE-to-technique mapping.

Dated corpus
The corpus is a focused snapshot and is not continuously refreshed.

Lexical grounding
Citations are checked lexically; full inference is out of scope.

Coverage score
The score measures evidence coverage. It does not measure whether conclusions are correct.

No expiry policy
Cached API records are not live lookups.

Limited concurrency
One API worker; close the database before replacing it.

Graph relations describe domain facts, not host compromise: PowerShell → subtechnique_of → T1059.

Teaching prototype only.